

MATH 526 Section 001 Lab 2 Spring 2006

Part I

In this part we learn to solve a lower triangular system by the method of forward substitution.

Create the following lower triangular matrix **A** and the vector **b**.

$$\mathbf{A} = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 4 & 5 & 2 & 0 \\ 3 & -2 & 1 & 1 \end{bmatrix} \quad \mathbf{b} = \begin{bmatrix} 2 \\ 1 \\ 6 \\ 4 \end{bmatrix}$$

The dimension of a matrix can be determined by the command **size**. Try the following.

```
>>[m,n]=size(A)
>>[m,n]=size(b)
```

Now we solve $\mathbf{Ax}=\mathbf{b}$ by forward substitution.

```
>>x(1)=b(1)/A(1,1)
>>x(2)=(b(2)-A(2,1)*x(1))/A(2,2)
>>x(3)=(b(3)-A(3,1)*x(1)-A(3,2)*x(2))/A(3,3)
>>x(4)=(b(4)-A(4,1)*x(1)-A(4,2)*x(2)-A(4,3)*x(3))/A(4,4)
```

The solution is now represented by a row vector. We can of course also represent the solution by a column vector

```
>>x=x'
>>clear x
```

This command removes the vector **x**.

Note that a product such as $\mathbf{A(3,1)*x(1)+A(3,2)*x(2)}$ can be expressed as the product of the vectors

$$[A(3,1) \ A(3,2)] \quad \text{and} \quad \begin{bmatrix} x(1) \\ x(2) \end{bmatrix}.$$

Hence we can replace $\mathbf{A(3,1)*x(1)+A(3,2)*x(2)}$ by $\mathbf{A(3,1:2)*x(1:2)'}.$ (Make sure you understand the meaning of all the terms. You saw them last week!)

We can therefore simplify the commands as follows.

```
>>x(1)=b(1)/A(1,1)
>>x(2)=(b(2)-A(2,1:1)*x(1:1)')/A(2,2)
>>x(3)=(b(3)-A(3,1:2)*x(1:2)')/A(3,3)
>>x(4)=(b(4)-A(4,1:3)*x(1:3)')/A(4,4)
```

Of course we can simplify further by using a simple **for** loop as follows. (Use the command **help for** for more information about the **for** loop.)

```
>>clear x
>>x(1)=b(1)/A(1,1);
>>for i=2:1:4
    x(i)=(b(i)-A(i,1:i-1)*x(1:i-1)')/A(i,i);
end
```

We have suppress the printing of the results of the computation. To see the answer, type

```
>>x=x'
```

We notice that the commands we used to solve $\mathbf{Ax}=\mathbf{b}$ are general commands, i.e., they can be applied to solve any 4×4 lower triangular system. So instead of typing the same commands every time we want to solve a 4×4 lower triangular system, we can create a file containing these commands and instruct MATLAB to execute the commands in the file.

Pop up an M-file window and type in the following lines.

```
function x=forward(A,b)
x(1)=b(1)/A(1,1);
for i=2:1:4
x(i)=(b(i)-A(i,1:i-1)*x(1:i-1)')/A(i,i);
end
x=x';
```

Save this file as forward.m Go back to your MATLAB window and type the following commands.

```
>>clear x
>>x=forward(A,b)
```

Matlab will execute the file forward.m using the matrix **A** and the vector **b** you have created before as the inputs and return the solution vector **x** as the output.

We can also write an M-file that will solve lower triangular systems of any dimension by using the command **size**. Modify your forward.m file as follows.

```

function x=forward(A,b)
[m,n]=size(A);
if m~=n, error('The input matrix is not a square matrix. '), end
x(1)=b(1)/A(1,1);
for i=2:1:m
x(i)=(b(i)-A(i,1:i-1)*x(1:i-1'))/A(i,i);
end
x=x';

```

Now you can solve any lower triangular system by using the M-file forward.m .

Question: What does the third line of forward.m do?

Let us create a 3×4 matrix **B**:

```
>>B=A(1:3,1:4)
```

Now we input **B** and **b** into forward.m:

```
>>x=forward(B,b)
```

What happened?

Use **help error** to find out more about the **error** command. Can you modify the M-file forward.m further so that it can also detect any discrepancy between the dimensions of A and b?

Before you start working on Part II, clear all the variables in your MATLAB window.